

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU annotator population is graduate/undergraduate students from US institutions [Q23] with Mechanical Turk and US 95th-percentile test-taker baselines [Q31, Q33, Q34] — no Arab regional stakeholders. Ground-truth labels reflect Western academic framing, with moral_scenarios explicitly encoding 'US standards as of 2020' (D12). Inter-annotator agreement is not documented. Empirical evidence (CAMEL, ILMAAM) confirms Western framing of Arab content in LLM evaluation systematically diverges from Arab stakeholder judgments [WEB-5, WEB-15]. With OC rated HIGH and the deployment serving eight Arab countries with contested historical narratives, this is a major convergent- and external-validity violation.

ID	Evidence
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
Q31	"Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test." (p.3)
Q33	"For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task." (p.3)
Q34	"If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-15	Naous et al. 2024 (CAMEL) empirically demonstrate LLM Western cultural bias in Arabic prompts (https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds)
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards

Strengths

- Annotator source institutions are documented [Q23, Q24, Q25], enabling diagnosis even though the demographics themselves are problematic
- Domain-expert performance is benchmarked against US 95th-percentile test-takers [Q33], providing a calibrated US-centric ceiling — useful as a baseline even if not regionally valid

ID	Evidence
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
Q24	"These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3)
Q25	"It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q33	"For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task." (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect Arab regional stakeholder perspectives. moral_scenarios labels are explicitly US-2020-anchored (D12); professional_law labels reflect US common law (D6–D8); us_foreign_policy labels reflect US institutional framing (D9–D11).

OC-2: High disagreement expected on contested topics (Palestinian narrative, Western Sahara, Jerusalem) where Arab stakeholder judgments diverge from US-academic-sourced labels [contested_topics YAML, WEB-5].

OC-3: MMLU paper does not document annotator demographics beyond 'graduate and undergraduate students' [Q23] — INSUFFICIENT DOCUMENTATION on country, ethnicity, or expertise composition.

OC-4: Re-annotation is what ArabicMMLU [WEB-2, WEB-6] and ILMAAM [WEB-5] effectively do — but for separate Arabic-content benchmarks, not for MMLU itself.

OC-5: INSUFFICIENT DOCUMENTATION — MMLU paper does not describe label aggregation methods or treatment of annotator disagreement; whether minority perspectives are erased cannot be assessed from the paper.

OC-6: Confirmed convergent- and external-validity violations: US-centric annotation [Q23, Q31, Q33], explicit US-2020 moral framing (D12), no Arab expert review documented; CAMEL evidence of LLM Western bias [WEB-15] confirms practical impact.

ID	Evidence
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure
D8	A buyer filed a lawsuit against a seller based on a written contract allegedly executed at the time of the sale of the seller's hot dog stand — US contract law
D9	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — framed entirely from US perspective
D10	In American government, the power to declare war rests with Congress. — US constitutional structure
D11	Revelations that the NSA was monitoring the communications of American citizens without obtaining warrants — US domestic surveillance scandal
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-15	Naous et al. 2024 (CAMEL) empirically demonstrate LLM Western cultural bias in Arabic prompts (https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds)
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-6	ArabicMMLU annotated by 10 Arab native speakers from Jordan, Egypt, Lebanon, UAE, KSA — illustrating the regional annotation MMLU lacks (https://github.com/mbzuai-nlp/ArabicMMLU)
Q31	"Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test." (p.3)

Q33	"For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task." (p.3)
Q79	"They are notably poor at modeling human (dis)approval, as evident by the low performance on the Professional Law and Moral Scenarios tasks." (p.8)

Information Gaps

- Annotator country/ethnicity composition beyond 'students'
- Inter-annotator agreement statistics (not reported in MMLU paper)
- Aggregation methodology for any items where annotators disagreed (not reported)

Requires Expert Verification

- Arab regional expert relabeling of any Arab-region or Islamic-content questions in MMLU (e.g., world_religions Islamic items, world_history Middle East items)

Remediation

Gap 1: Ground-truth labels were determined by US graduate/undergraduate annotators with no documented Arab regional stakeholder review.

Recommendation: For any MMLU items retained that touch Arab or Islamic content (world_religions Islamic items, world_history Middle East items, geography items), commission Arab regional expert relabeling. Use ILMAAM's 11-expert annotation methodology [WEB-5] as a template.

ID	Evidence
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)